

Antelligence!

The idea of large groups of simple nanobots acting together, under the influence of their collective intelligence, leads to exciting as well as dire prospects

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*Go to the ant, you sluggard;
Consider its ways and be wise!
It has no commander,
No overseer or ruler,
Yet it stores its provisions in summer
And gathers its food at harvest.*

—Proverbs 6:6

Google “grey goo”. You’ll find the definition, one of which is “Gray goo is a mass of small, destructive, self-replicating nanorobots.” But amongst other things, you’ll find, in the results, “fear of runaway robots”; “the grey goo threat”; “worse than grey goo”; and yes, even “the end of the world may be nearer than you think.” Clearly, grey goo is not a thing you’d like to be associated with.

This variety of *goo*, according to some, will be the way the world ends. We’ll have more to say about grey goo later; for now, we’re hoping that swarms of nanorobots will remain under control and do our bidding—exploring planets, repairing arteries, and other noble deeds.

So how did we get from ants to nanobots—robots so tiny they’re measured on the nanoscale? It’s scientific, as you’ll find out. Let’s get back to the ant.

Having no commander, the ant yet gathers its food. This is a marvel of nature: does an ant even have a brain? Even if it does, it doesn’t go a long way in helping the ant gather its food. What helps the ant more is the fact that it is part of a larger entity called the hive. *The*

hive exhibits intelligence that an individual ant is incapable of by itself—the whole is greater than the sum of its parts.

Philosophy alert! Did we just touch upon a contentious issue? We did, apparently; we can already hear the clamours from those who disagree. Yes, wholes and holism and reductionism are touchy subjects, and we will not veer where many philosophers have gone before.

Back to the ant again: our glorification of that creature is not to imply that the bee is far behind. Each individual bee is reasonably dumb, but a hive of them can be dangerously co-operative. This we naïvely attribute to “collective intelligence”. We are re-stressing our point, but the concept is essential. And “It turns out that what makes sense in the biological world often makes sense in the computational world as well,” confirms Christian Jacob, leader of the Evolutionary and Swarm Design Research Group at the University of Calgary, Canada.

So what does all this have to do with computers? Not much as graphics cards go, but it has a lot to do with the idea of swarm intelligence (SI). If you like formal definitions, SI is the property of a system whereby the collective behaviours of unsophisticated agents interacting locally with their environment cause coherent functional global patterns to emerge. In English, SI is the idea that a large number of relatively stupid things can together exhibit useful, intelligent behaviour.

This is a story about how the group behaviour of insects and other fauna led scientists to dream of expansive swarms of tiny robots doing useful work.

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